Supplementary Material for: Temporal decoupling of CTLA-4 and ERK sensitivity in brain tumour signaling: a theoretical basis for sequencing checkpoint and MAPK inhibition

This file includes equations that support the main manuscript.

Supplementary file S1: ODE Equations

Table S1. System of ordinary differential equations (ODEs) governing the CTLA-4-mediated signaling pathway.

Eq. No.	Equation
1	$\frac{d[\text{CTLA-4}]}{dt} = -\Big(k^{+}_{\text{CTLA4_CD80}}[\text{CTLA-4}][\text{CD80}] - k^{-}_{\text{CTLA4_CD80}}[\text{CTLA4_CD80}]\Big) - k^{+}_{\text{CTLA4_MDX-CTLA4}}[\text{CTLA-4}][\text{MDX-CTLA4}]$
2	$\frac{d[\text{CD80}]}{dt} = -\Big(k^{+}_{\text{CTLA4_CD80}}[\text{CTLA-4}][\text{CD80}] - k^{-}_{\text{CTLA4_CD80}}[\text{CTLA4_CD80}]\Big)$
3	$\frac{d[\text{CTLA4_CD80}]}{dt} = k^{+}_{\text{CTLA4_CD80}}[\text{CTLA-4}][\text{CD80}] - k^{-}_{\text{CTLA4_CD80}}[\text{CTLA4_CD80}]$
4	$\frac{d[\text{RTKs}]}{dt} = -\Big(k^{+}_{\text{RTKs_Grb2}}[\text{RTKs}][\text{Grb2}] - k^{-}_{\text{RTKs_Grb2}}[\text{RTKs_Grb2}]\Big) - k^{+}_{\text{PI3K_RTKs}}[\text{RTKs}][\text{PI3K}] - k^{+}_{\text{RTKs_NF2}}[\text{RTKs}][\text{NF2}] - k^{+}_{\text{RTKs_GW572016}}[\text{GW572016}][\text{RTKs}]$
5	$\frac{d[\text{Grb2}]}{dt} = -\Big(k^{+}_{\text{RTKs_Grb2}}[\text{RTKs}][\text{Grb2}] - k^{-}_{\text{RTKs_Grb2}}[\text{RTKs_Grb2}]\Big)$
6	$\frac{d[\text{RTKs_Grb2}]}{dt} = \Big(k^{+}_{\text{RTKs_Grb2}}[\text{RTKs}][\text{Grb2}] - k^{-}_{\text{RTKs_Grb2}}[\text{RTKs_Grb2}]\Big) - \Big(k^{+}_{\text{RTKs_Grb2_SOS}}[\text{RTKs_Grb2}][\text{SOS}] - k^{-}_{\text{RTKs_Grb2_SOS}}[\text{RTKs_Grb2_SOS}]\Big)$
7	$\frac{d[\text{SOS}]}{dt} = -\Big(k^{+}_{\text{RTKs_Grb2_SOS}}[\text{RTKs_Grb2}][\text{SOS}] - k^{-}_{\text{RTKs_Grb2_SOS}}[\text{RTKs_Grb2_SOS}]\Big)$

Eq. No.	Equation
8	$\frac{d[\text{RTKs_Grb2_SOS}]}{dt} = \Big(k^{+}_{\text{RTKs_Grb2_SOS}}[\text{RTKs_Grb2}][\text{SOS}] - k^{-}_{\text{RTKs_Grb2_SOS}}[\text{RTKs_Grb2_SOS}]\Big) - k^{+}_{\text{RasGTP_RTKs_Grb2_SOS}}[\text{RTKs_Grb2_SOS}]$
9	$\frac{d[\text{Ras-GTP}]}{dt} = k^{+}_{\text{RasGTP_RTKs_Grb2_SOS}}[\text{RTKs_Grb2_SOS}] - k^{+}_{\text{Raf_RasGTP}}[\text{Ras-GTP}][\text{Raf}] - k^{+}_{\text{RasGTP_PP2A}}[\text{Ras-GTP}][\text{PP2A}] - k^{+}_{\text{RasGTP_NF2}}[\text{Ras-GTP}][\text{NF2}] - k^{+}_{\text{PI3K_RasGTP}}[\text{Ras-GTP}]$
10	$\frac{d[\text{Raf}]}{dt} = k^{+}_{\text{Raf_RasGTP}}[\text{Ras-GTP}][\text{Raf}] - k^{+}_{\text{MEK_Raf}}[\text{Raf}] - k^{+}_{\text{Raf_PP2A}}[\text{Raf}][\text{PP2A}]$
11	$\frac{d[\text{MEK}]}{dt} = k^{+}_{\text{MEK_Raf}}[\text{Raf}] - k^{+}_{\text{ERK_MEK}}[\text{MEK}][\text{ERK}] - k^{+}_{\text{MEK_PP2A}}[\text{MEK}][\text{PP2A}]$
12	$\frac{d[\text{ERK}]}{dt} = k^{+}_{\text{ERK_MEK}}[\text{MEK}][\text{ERK}] - k^{+}_{\text{CyclinD_ERK}}[\text{ERK}] - k^{+}_{\text{p16_ERK}}[\text{ERK}][\text{p16}] - k^{+}_{\text{ERK_PP2A}}[\text{ERK}][\text{PP2A}] - k^{+}_{\text{ERK_BVD523}}[\text{BVD-523}][\text{ERK}]$
13	$\frac{d[\text{PI3K}]}{dt} = k^{+}_{\text{PI3K_RTKs}}[\text{RTKs}][\text{PI3K}] - k^{+}_{\text{PDK1_PI3K}}[\text{PI3K}][\text{PDK1}] + k^{+}_{\text{PI3K_SMO}}[\text{SMO}] - k^{+}_{\text{PI3K_PTEN}}[\text{PI3K}][\text{PTEN}] + k^{+}_{\text{PI3K_RasGTP}}[\text{Ras-GTP}]$
14	$\frac{d[\text{AKT}]}{dt} = -k^{+}_{\text{mTOR_AKT}}[\text{AKT}] + k^{+}_{\text{AKT_PDK1}}[\text{PDK1}][\text{AKT}] - k^{+}_{\text{AKT_PP2A}}[\text{AKT}][\text{PP2A}]$
15	$\frac{d[\text{mTOR}]}{dt} = k^{+}_{\text{mTOR_AKT}}[\text{AKT}] - k^{+}_{\text{mTOR_NF2}}[\text{mTOR}][\text{NF2}]$
16	$\frac{d[\text{SMO}]}{dt} = -k^{+}_{\text{PI3K_SMO}}[\text{SMO}]$
17	$\frac{d[\text{PP2A}]}{dt} = \frac{K^{\text{PP2A}}_{\text{cat,CTLA4_CD80}}\,[\text{CTLA4_CD80}]^{n_1}}{K^{\text{PP2A}}_{\text{m,CTLA4_CD80}} + [\text{CTLA4_CD80}]^{n_1}} - k^{+}_{\text{PP2A_ERK}}[\text{PP2A}][\text{ERK}]$
18	$\frac{d[\text{CyclinD}]}{dt} = -\Big(k^{+}_{\text{CyclinD_CDK4}}[\text{CyclinD}][\text{CDK4}] - k^{-}_{\text{CyclinD_CDK4}}[\text{CyclinD_CDK4}]\Big) + k^{+}_{\text{CyclinD_ERK}}[\text{ERK}]$

Eq. No.	Equation
19	$\frac{d[\text{CDK4}]}{dt} = -\Big(k^{+}_{\text{CyclinD_CDK4}}[\text{CyclinD}][\text{CDK4}] - k^{-}_{\text{CyclinD_CDK4}}[\text{CyclinD_CDK4}]\Big) - k^{+}_{\text{CDK4_p16}}[\text{CDK4}][\text{p16}]$
20	$\frac{d[\text{RB}]}{dt} = k^{+}_{\text{RB_CyclinD_CDK4}}[\text{CyclinD_CDK4}] - k^{+}_{\text{RB_CyclinE_CDK2}}[\text{RB}][\text{CyclinE_CDK2}]$
21	$\frac{d[\text{E2F}]}{dt} = -k^{+}_{\text{G1-S_E2F}}[\text{E2F}] - k^{+}_{\text{CyclinE_E2F}}[\text{E2F}] - k^{+}_{\text{p14_E2F}}[\text{E2F}] + \frac{K^{\text{E2F}}_{\text{catRB}}[\text{RB}]^{n_2}}{K^{\text{E2F}}_{\text{mRB}} + [\text{RB}]^{n_2}}$
22	$\frac{d[\text{G1-S}]}{dt} = k^{+}_{\text{G1-S_E2F}}[\text{E2F}] - k^{+}_{\text{G1-S_p21}}[\text{G1-S}][\text{p21}]$
23	$\frac{d[\text{p14}]}{dt} = k^{+}_{\text{p14_E2F}}[\text{E2F}] - k^{+}_{\text{p14_p53}}[\text{p14}][\text{p53}]$
24	$\frac{d[\text{MDM2}]}{dt} = -k^{+}_{\text{MDM2_p14}}[\text{MDM2}][\text{p14}] + \frac{K^{\text{MDM2}}_{\text{catp53}}[\text{p53}]^{n_3}}{K^{\text{MDM2}}_{\text{mp53}} + [\text{p53}]^{n_3}} - k^{+}_{\text{MDM2_AMG232}}[\text{MDM2}][\text{AMG-232}]$
25	$\frac{d[\text{p53}]}{dt} = -k^{+}_{\text{p21_p53}}[\text{p53}] - k^{+}_{\text{p53_MDM2}}[\text{p53}][\text{MDM2}]$
26	$\frac{d[\text{p21}]}{dt} = k^{+}_{\text{p21_p53}}[\text{p53}]$
27	$\frac{d[\text{CyclinD_CDK4}]}{dt} = k^{+}_{\text{CyclinD_CDK4}}[\text{CyclinD}][\text{CDK4}] - k^{-}_{\text{CyclinD_CDK4}}[\text{CyclinD_CDK4}]$
28	$\frac{d[\text{p16}]}{dt} = k^{+}_{\text{p16_ERK}}[\text{ERK}][\text{p16}]$
29	$\frac{d[\text{CyclinE}]}{dt} = k^{+}_{\text{CyclinE_E2F}}[\text{E2F}] - \Big(k^{+}_{\text{CyclinE_CDK2}}[\text{CDK2}][\text{CyclinE}] - k^{-}_{\text{CyclinE_CDK2}}[\text{CyclinE_CDK2}]\Big) - k^{+}_{\text{CyclinE_p27}}[\text{CyclinE}][\text{p27}]$

Eq. No.	Equation
30	$\frac{d[\text{CDK2}]}{dt} = -\Big(k^{+}_{\text{CyclinE_CDK2}}[\text{CDK2}][\text{CyclinE}] - k^{-}_{\text{CyclinE_CDK2}}[\text{CyclinE_CDK2}]\Big) - k^{+}_{\text{CDK2_p21}}[\text{CDK2}][\text{p21}]$
31	$\frac{d[\text{CyclinE_CDK2}]}{dt} = k^{+}_{\text{CyclinE_CDK2}}[\text{CDK2}][\text{CyclinE}] - k^{-}_{\text{CyclinE_CDK2}}[\text{CyclinE_CDK2}]$
32	$\frac{d[\text{p27}]}{dt} = -k^{+}_{\text{p27_AKT}}[\text{p27}][\text{AKT}]$
33	$\frac{d[\text{PDK1}]}{dt} = k^{+}_{\text{PDK1_PI3K}}[\text{PI3K}][\text{PDK1}] - k^{+}_{\text{AKT_PDK1}}[\text{PDK1}][\text{AKT}]$